Removal Sample Audit: Initiation document · Case 03

Following chapter 1 (Initiation) of the Magshimim project-book template.

Description

A system that measures how many deletions from the electoral roll were wrong without checking all of them. In each district it draws a random sample of a few hundred deletions, sends each case for a full check (does the person still live here, are they alive, are they registered once), and estimates the district-wide share of wrong deletions with a known margin of error.

Instead of a political argument about "how many were removed by mistake", each district gets one checkable number: "11% ±1.8 of deletions here are wrong". A district with a high number gets a full re-check before the election.

Goal

Two camps look at the same 130 million and conclude opposite things. Both are partly right and nobody knows how much. The only hard signal, that most decided appeals succeeded, suggests the error is not small. Goal: replace the argument with a measurement that is old, cheap, understood and fast enough to run before the polls.

Audience

- **Primary:** the Election Commission and the courts, who must decide whether a revision is reliable enough to hold an election on.
- **Secondary:** parties, journalists and researchers who want a number they can reproduce.
- **Scenario:** with 37 lakh appeals pending in West Bengal, the ECI audits 1,500 deletions per district in two weeks and learns which districts need a full stop-and-check.

Existing solutions

- **Risk-limiting audits (US):** statistical ballot audits used by several US states; samples are accepted as a legal basis for trust, but for the count, not the roll.
- **Civil-society roll audits in India:** one-off field sample checks by groups such as ADR and independent researchers, without a standard method or status.
- **ECI's own house-to-house verification by booth-level officers:** exists, but is not a random sample and publishes no margin of error.

What is new: a reproducible random draw from a public seed, stratified by deletion reason, with method, seed and results published.

Technological core

The sampling and estimation engine: a verifiable random draw from a public seed, stratification by reason, and a Wilson confidence interval per district and reason.

Main technologies

Python with NumPy and SciPy; a public randomness beacon for the seed; an offline-capable Android field app; PostgreSQL and Superset for results. Open questions: simple versus stratified sampling; field visit versus phone check.

Challenges and sources

- **Ground truth:** whether a person still lives here is known only from a visit. Who visits, and whom do we trust? This is the central challenge.
- **Access to the deletion list:** needed to draw the sample; after the 2025 Bihar order such lists were published.
- **Scale:** hundreds of districts × 1,500 cases = hundreds of thousands of visits; large, but far smaller than 130 million.

Sources: Stark, "Risk-Limiting Postelection Audits"; Cochran, Sampling Techniques; Supreme Court order on Bihar (August 2025).

Hardware

None special.

Conclusion

✅ Survives. Old, accepted method; public input; clear output. Condition: ground truth. Research must test how sensitive the estimate is to field checkers who do not really check.